Global–Local Feature Collaborative Fusion and Edge Refinement Network for Remote Sensing Change Detection

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***Abstract*— The remote sensing image change detection (RSI-CD) has witnessed significant progress. The core focus lies in accurately identifying spatiotemporal changes in remote sensing images and precisely extracting edge details. However, the current remote sensing image change detection (CD) primarily faces two challenges: the diversity and complexity of change regions and the blurred edges of change regions. These issues lead to pseudo-change interference and incomplete detection of change targets. To alleviate these limitations, this study first develops a local–global differential fusion module (LGDFM) that integrates local feature extraction with global self-attention mechanism to enhance change feature representation through the effective bitemporal image fusion. This module simultaneously captures both global and local information from source images, thereby overcoming the insufficient feature fusion in existing bitemporal CD methods. Subsequently, we propose an edge-aware enhancement module (EAEM) that employs morphological operations to extract edge features. The module progressively refines prediction results through multistage edge feature integration that ensures precise contour delineation. Experiments on two datasets demonstrate that the proposed method achieves competitive performance compared with existing approaches.**

***Index Terms*— Change detection (CD), differential fusion, edge-aware enhancement, remote sensing images.**

I. Introduction

CHANGE detection (CD) in remote sensing is defined as the identification of alterations in land cover types over different time intervals by analyzing two or more remote sensing images of the same geographical area captured at different times. However, the increasing resolution of remote sensing images poses significant challenges to the detection task due to the diversity of changing regions and edge blurring issues. Unlike traditional monitoring methods, it effectively captures dynamic changes on the Earth's surface through the analysis of large-scale and high-resolution remote sensing data. This capability renders CD crucial in various fields, such as Earth's science [1], environmental monitoring [2], urban planning [3], and disaster response [4].

In traditional CD research, mainstream methods primarily rely on manually designed features and rule-based approaches. While these methods perform well in specific datasets and scenarios, their limitations become evident as the resolution of remote sensing images increases and the scenes grow more complex. The traditional CD algorithms mainly comprise two typical categories: pixel-level-based analysis methods and object-level-based processing approaches [5]. Pixel-based CD performs change identification by directly comparing corresponding pixel values between two images [6]. However, this approach demonstrates limited effectiveness when applied to cross-sensor or cross-temporal image pairs [7]. In contrast, object-based CD methods focus on analyzing changes at the object level after segmentation. However, traditional methods struggle to accurately detect changed regions when dealing with complex land-cover changes.

Deep learning-based CD methods [8] are more diverse in classification compared with traditional methods. CNN-based CD methods primarily employ CNN architectures to achieve feature encoding and decoding of remote sensing images. Their core advantage lies in the efficient extraction of local spatial features, such as edges and textures. These methods typically adopt an encoder–decoder structure, where the encoder gradually extracts deep features through multilayer convolutions and downsampling. The decoder achieves precise localization [9] of change regions through upsampling and feature fusion. For example, the network proposed by Fang et al. [10] employs a dense skip connection between encoder and decoder. This architecture not only effectively preserves high-resolution features and accurately localizes details but also introduces a densely connected structure within the decoder. However, the local receptive field of CNNs inherently limits their ability to capture global information. To partially address this issue, some studies integrate self-attention mechanisms [11], [12], [13], [14] into CNN models, mitigating the limitations of the receptive field to some extent. Subsequently,

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research has focused on incorporating attention mechanisms to further enhance change information extraction [15], [16], [17]. For instance, Li et al. [18] combined a neighborhood feature attention module (NFAM) with a semantic filtering mechanism. This approach significantly raises the accuracy of CD by enhancing the semantic representation and filtering pseudo-change. However, these CNN-based methods often struggle with capturing long-range dependencies due to their limited receptive fields, which can lead to incomplete detection of large-scale changes in complex scenes.

Transformer-based CD methods deeply integrate Transformer technology into the encoder or decoder modules. By leveraging its unique self-attention mechanism and multihead attention structure, these methods can effectively model long-range dependencies in remote sensing images. Compared with the limited local receptive fields of traditional convolutional neural networks (CNNs), Transformers significantly enhance the ability of the model to capture large-scale change features in multitemporal images through their global context modeling capability. In addition, existing approaches have proposed hybrid CD methods [19], [20], aiming to fuse the technical advantages of CNN and Transformers. These methods typically extract bitemporal change features through CNN and Transformer branches during the encoding stage and then reconstruct change regions through decoding operations [21], [22]. For example, Feng et al. [23] proposed the Intra-Scale Cross-Interaction and Inter-Scale Feature Fusion Network (ICIF-Net). The method adopts a parallel dual-branch architecture with CNN and Transformer, integrating local and global features through interaction and fusion modules. A feature difference network then generates feature difference maps. The approach significantly improves the performance of CD. Meanwhile, Tang et al. [24] developed the W-shaped hierarchical network (WNet) model by leveraging the respective strengths of CNN and Transformers. This hybrid architecture can both precisely extract local detailed features and effectively model global contextual information, ultimately achieving significant performance improvements. Despite their advantages, Transformer-based methods may overlook fine-grained local details, and hybrid approaches often face challenges in effectively balancing the tradeoff between global context modeling and local feature extraction, resulting in suboptimal performance for targets with intricate boundaries.

In summary, although existing methods for remote sensing image CD have achieved some results, these approaches generally have relatively limited exploration in feature fusion. In addition, edge detail detection in images is crucial for CD tasks [25]. Edge information [26] often contain important features, such as contours, shapes, and structures of targets, which tend to be particularly prominent during the change process. Traditional CD methods often overlook this edge information, resulting in poor performance when dealing with scenes featuring complex structures and irregular changes. Therefore, designing a CD method that effectively fuses dual-temporal features and accurately extracts edge information becomes a focal point of current research. To this end, this article proposes a global–local feature collaborative fusion and edge-aware refinement network. The method consists of three components: a local–global fusion module, an edge refinement module, and a decoder. Each module explores relevant knowledge from a unique perspective, thereby improving CD accuracy. The contributions of the proposed approach can be described as follows.

1) To alleviate the issues of missed small targets and poor adaptation to complex scenarios caused by single-scale feature inputs, we propose a local–global differential fusion module (LGDFM). This module employs a feature difference mechanism to jointly model local–global discrepancies. It adaptively balances the contributions of local and global features, thereby capturing the semantic information of images.
2) The edge-aware enhancement module (EAEM) optimizes the contour accuracy of change targets through a morphological edge-aware mechanism. It applies dilation–erosion operations to coarse prediction results to explicitly extract edge features. By incorporating the long-range dependency modeling capability of the Transformer layer, it corrects geometric distortions caused by the local processing in CNNs.
3) We propose a remote sensing image CD method named global–local optimization network (GLONet), which seamlessly integrates the carefully designed LGDFM with EAEM. Experiments on two datasets demonstrate the better performance of our method in remote sensing image CD. Our approach outperforms several existing methods in this task.

This article is organized as follows. Section II reviews related works. Section III details the proposed GLONet method. Section IV evaluates experiments and ablation studies. Section V concludes this article.

II. Related Works

Remote sensing image CD has an extensive history, and we briefly review the CNN-based CD models as well as the dual-temporal feature fusion strategies, which form the foundation of our research.

A. CNN-Based CD Methods

Most CD methods based on CNN adopt the encoder–decoder architecture as the core framework. This architecture typically includes a Siamese convolutional encoder with shared weights. This architecture can hierarchically capture fused features, incorporating both spatial details and semantic information from time-series remote sensing images [27]. The decoder part gradually restores the spatial resolution through upsampling operations and combines mechanisms such as skip connections to fuse shallow detail features. Finally, a pixel-level classifier converts the change perception representation into a binary or multiclass change probability map. For example, Zhang and Shi [28] proposed a CD method based on FDCNN. The method utilizes the VGG16 network to learn deep features and finally employs a thresholding approach to generate the final change binary map. On this basis, the multiscale characteristics learning strategy [10], [29], [30] is an important method to improve the context modeling ability

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of deep neural networks. The core concept of this method lies in integrating feature representations across different levels and scales, thereby effectively enhancing the generalization of model capability in complex scenarios. Even with these technological advancements, CD in complex scenes yet faces several challenges: the scale diversity of target objects in large-scale scenes imposes higher demands on feature extraction, while the inherent background complexity of images makes the precise delineation of change boundaries particularly difficult. These factors collectively limit the performance of existing methods in practical applications.

B. Dual-Temporal Feature Fusion Strategies

Currently, deep learning-based CD methods primarily employ a dual-stream encoder architecture as the mainstream approach for processing dual-temporal images. This type of architecture extracts multiscale deep features of images at different time phases through parallel branches, respectively. However, its performance bottleneck often lies in the interaction and fusion of features across time phases. Specifically, the cross-scale dynamic fusion module proposed in the literature [31] significantly improves the CD accuracy by improving the feature fusion method. Similarly, the attention-guided feature interaction mechanism developed in the literature [32] also significantly enhances the CD accuracy in the same way. They verify the core position of the feature fusion strategy in the dual-stream network. In addition, the existing traditional fusion strategies show obvious limitations when dealing with CD tasks. To this end, existing methods propose that, for example, Ma et al. [33] fused the dual-temporal features through a cross-time gating mechanism and a cross-scale attention mechanism, respectively, to highlight the changes of interest. Wang et al. [34] proposed a feature interleaved fusion module to effectively fuse the features of dual-temporal images. The fusion strategy has made progress at the current stage. However, existing methods often find it difficult to effectively balance the global context information and local detail features during the fusion process, which restricts the expressive ability of change features. Therefore, this article proposes a global–local collaborative feature fusion module, which achieves efficient differential fusion of dual-temporal features by jointly modeling long-range dependency relationships and local variation features. In addition, we propose the EAEM method to alleviate the challenge of obtaining complete boundaries to enhance the extraction of image boundaries.

III. Proposed Method

A. Overview

For CD tasks, we face challenges, such as image discrepancies, diverse and complex change regions, and blurred edges of change regions. These factors make the detection task more complex compared with traditional CD. Although a significant amount of research has been devoted to alleviating these issues, many methods still struggle to accurately extract details of change regions. This shortcoming is primarily due to insufficient attention to three key aspects: feature extraction, feature difference, and edge details. Specifically, it is crucial for precise guidance modules to focus on change regions to extract targets. In addition, it is essential to preserve the inherent characteristics of change regions throughout the process to promote effective feature extraction. In summary, when designing models for CD, thorough consideration of identifying, extracting change regions, and edge details is crucial. This approach will contribute to achieving more accurate detection results.

To alleviate the challenges in existing methods, such as the diversity and complexity of changing regions, as well as the blurred edges of change areas, we propose a GLONet for CD. The overall flowchart is shown in Fig. 1. The network consists of a feature extraction layer, difference enhancement, edge refinement, and a decoder. Each part addresses the CD problem from a different perspective. In feature extraction, we employ a standard shared-weight Siamese ResNet-18 model as the backbone network. Considering the diversity of image change regions, we propose a LGDFM to enhance the ability of model to capture change features through the differential fusion of dual-temporal image features. Unlike conventional single-scale or fixed multiscale approaches, the GDFM module achieves adaptive interaction between GFEM and LFEM through feature difference module (FDM), jointly modeling both local and global discrepancies. To address the issue of blurred edges in change regions observed in existing methods, an EAEM is designed to extract edge features and refine the prediction results. In addition, a self-attentive feature guidance (SAFG) map with global semantic information is generated using deep high-level semantic information and a self-attention mechanism. This SAFG map is integrated into the MSFA module to guide the aggregation of multilevel semantic information. Meanwhile, the MSFA module employs multiscale convolutional operations to characterize fine-grained change features, enhancing the complete reconstruction of change targets. Finally, a convolutional module is used to classify changed and unchanged pixels.

B. Local–Global Differential Fusion Module

Considering that CD tasks typically involve processing bitemporal images, which may exhibit pseudo-change in certain areas due to differences in acquisition conditions (e.g., illumination, season, or sensor variations), we propose a local–global difference fusion module for feature fusion and difference computation of images. As illustrated in Fig. 2, the module first employs a dual-path extraction mechanism to separately capture local details and global contextual information from the images. Then, it computes feature differences between the extracted features to generate difference features.

1) GFEM: Global context helps the model identify changes between different regions and is key to distinguishing various types of changes [35]. In order do this, we propose the GFEM, which extracts global features through a series of operations. Considering that change features at different scales often exist in bitemporal images, the module employs convolutional layers to effectively capture local and medium-scale features. Meanwhile, 1×1 convolutional layers integrate global information across different levels, preserving the integrity of the global structure. To ensure that basic information is

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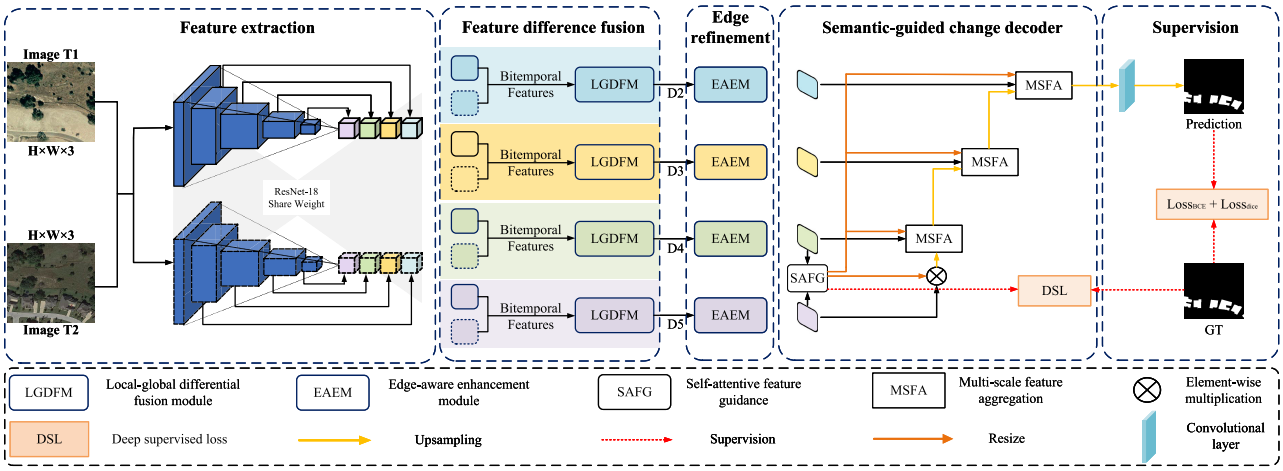


Fig. 1. Illustration of the implementation process of GLONet.

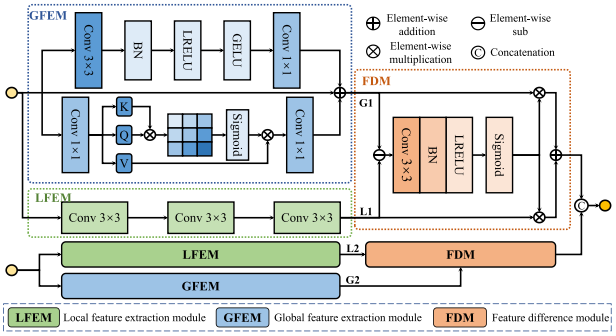


Fig. 2. Illustration of the implementation process of LGDFM.

not lost while fusing deeply extracted change features, skip connections are introduced in the module. In addition, changes in regions may be distributed across different parts of the image. The importance of change information in different regions varies for the final detection result. The model incorporates an attention mechanism to dynamically enhance feature weights in change regions while suppressing interference from nontarget areas. It is able to adaptively focus on change regions in the image, thereby improving the recognition capability for significant change areas.

2) Feature Difference Module: Image variation is typically manifested as feature differences in the target region. Directly inputting the original images into a difference model may overlook significant changes between the two images. To better capture variations between images, FDM is introduced to mitigate feature inconsistency issues. Considering that image changes may reflect differences in various regions, we integrate convolutional layers with different operations. Convolutional layers extract detailed features from the images, while difference operations help identify significant change regions between the two images. Since image variations often involve subtle feature differences, activation functions are employed to enhance the expressive power of these differences. In addition, to effectively capture changes, elementwise multiplication is performed on the input features. This elementwise multiplication operation emphasizes change information at the pixel level, suppresses irrelevant regions, and enhances the saliency of image variations. Finally, the feature maps from GFEM and LFEM are introduced into the FDM and fused to produce the final feature map. The GFEM excels at modeling long-range dependencies. Conversely, the LFEM preserves spatial details effectively. Through a sigmoid function, both GFEM and LFEM generate channelwise weights to adaptively adjust the contribution of their respective features in the fusion process. The final fused feature is expressed as follows:

$$F_{\text{fused}} = F_l \cdot \Delta F + F_g \cdot (1 - \Delta F) \tag{1}$$

where ΔF is an adaptive weight map, F_l is the local features, and F_g is the global features.

C. Edge-Aware Enhancement Module

Edge information often contain important features (such as contours and shapes), to enhance edge detail information, we propose the EAEM. The module architecture is depicted in Fig. 3. The entire process extracts local features through convolution, while dilation and erosion operations emphasize structural information in the image. Elementwise multiplication then helps highlight significant changing regions or features. Finally, a Transformer encoder captures global contextual information to complete edge feature extraction. Notably, the difference feature map generated by LGDFM is directly fed into the EAEM as input.

During the edge enhancement [36] process, the input feature size is reshaped to fit the subsequent convolutional operations, which is depicted as follows:

$$F = C_{\text{BR}}(C_{\text{BR}}(C_{\text{BR}}(R(X_i)))) \tag{2}$$

where $F \in \mathbb{R}^{H\times W\times C}$, $C_{\text{BR}}(\cdot)$ represent a convolutional layer, normalization, and ReLU operations. $R(\cdot)$ represents the reshaping operation. Then, the input features are further extracted through three layers of Conv–BN–ReLU combination, which gradually enhances the feature representation capability. This multilayer stacked construction better extracts detailed and semantic information from input features. It provides richer feature representations for subsequent prediction

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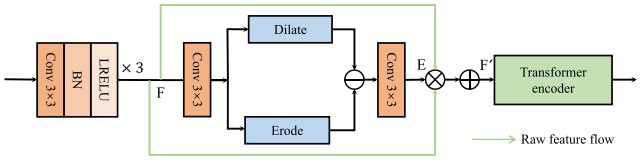


Fig. 3. Illustration of the implementation process of EEAM.

and edge extraction. During the subsequent edge extraction process, the edge details are often not precisely captured. To address this, we first employ a convolutional layer to obtain coarse prediction features, which initially localize the salient regions in the input image. Through rough prediction, the model can roughly determine the position and shape of the salient objects. Subsequently, the edge characteristics E are obtained by applying dilation and erosion operations to the low-resolution prediction outputs. They are expressed as follows:

$$E = C_{3\times3}\left(\text{dilate}\left(P_i^1\right) - \text{erode}\left(P_i^1\right)\right) \tag{3}$$

where $E \in \mathbb{R}^{H\times W\times C}$, dilate$(\cdot)$, and erode$(\cdot)$, respectively, represent the expansion and erosion operation, and $C_{3\times3}(\cdot)$ expresses a 3×3 convolutional layer.

To enhance the edge detail information of features, it is proposed to multiply and add the edge feature E and the feature F pixel-by-pixel. By multiplying pixel-by-pixel, the edge information is weighted with the original features, highlighting the details of the edge regions. Subsequently, the weighted edge information is fused with the original features through the addition operation, thereby further enhancing the edge details while preserving the original features. This operation is to fuse the edge information with the original features, enabling generation of clearer edges in salient object detection, resulting in F'. Traditional CNNs are limited by their local receptive fields that is difficult for them to capture global semantic information in images. Especially in complex scenes, the local receptive fields in CNNs limit their ability to maintain distant feature correlations. In addition, the blurring of local regions (such as unclear edges or loss of details) is a common challenge in salient object detection. Traditional CNN models may not effectively restore detailed information when processing these blurred regions. To address this, it is proposed to unfold the refined feature F' and feed it into a Transformer layer. The local features extracted by CNN and the global semantic features captured by Transformer are synergistically integrated to achieve a more comprehensive feature representation.

D. Semantic-Guided Change Decoder

Current CD methods reveal that the transition between low-level and high-level features is not smooth and continuous, but rather exhibits a leap that cannot be directly correlated. Direct aggregation of multiscale features would impair the representational effectiveness of hierarchical features. To mitigate this issue, we adopted the decoder design from [37]. Specifically, it integrates the last-layer and penultimate-layer features with a self-attention mechanism to generate the SAFG map. The semantic guidance provides prior information for multiscale feature fusion through components such as the SAFG map, ensuring the decoding process focuses on genuine change regions while suppressing pseudo-change interference. During the decoding phase, the SAFG map is embedded into the multiscale feature aggregation (MSFA) module, which combines multiscale convolutional operations to produce the final prediction results.

The SAFG module integrates deep features from the last two layers through a self-attention mechanism to generate a SAFG. This map encodes global semantic relationships of change regions and serves as prior knowledge to guide the fusion of multiscale features in the decoder. It approximates the real change target through deep supervised loss to provide more accurate guidance. Specifically, the module starts by inputting the last layer and the previous layer features. Considering the differences in spatial and channel dimensions between the two features, we adopt upsampling and a convolutional layer to align the spatial scales and compress the feature channels of the high-level features. Next, the aligned features are concatenated and their representations are enhanced through a convolutional layer. Since convolutional layer typically capture only local neighborhood feature relationships and fail to effectively model dependencies between distant pixels, they lack sufficient global contextual information. Therefore, a self-attention mechanism is introduced to the convolutional features in subsequent operations. The mechanism captures global contextual information by computing relationships between all positions in the feature map. It can reduce the impact of irrelevant details. Moreover, to effectively preserve original feature information and avoid the gradient vanishing problem in deep networks, we put forward a skip connection from the features of the penultimate layer and add it to the self-attention output. Finally, the combined features are processed to generate the probability map.

The MSFA aggregates multilevel semantic information through the SAFG map. Since the channel dimensions of features at different levels vary, an upsampling operation is first applied to align the spatial scales of the two sets of features, followed by concatenation along the channel dimension. Subsequently, the obtained SAFG map is used as prior information to weight the concatenated multilevel features. In the subsequent operations, convolutional layers are used to reduce the channel dimension, compressing the rich information in the high-dimensional feature space into a more compact form. Considering that change targets in remote sensing images may vary in size and shape (e.g., buildings, roads, and vegetation), we adopt a multiscale convolutional layer to extract fine-grained multiscale change features. By processing features at different scales in parallel, multiscale convolution can more comprehensively describe the geometric structure and boundary information of change targets. Due to the limited sensory field of ordinary convolution and the difficulty of capturing long-range dependencies, while considering the omnidirectional nature of change regions in practical applications, three dilated convolutions with different dilation rates are employed for feature extraction in our method. Dilated convolutions can expand the receptive field while maintaining feature map resolution, which helps accurately localize

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change boundaries. Finally, the prediction results are generated through a convolutional layer, normalization, and an activation function.

E. Loss Function

After completing the model construction, we need to carry out model training and evaluation. We construct a hybrid loss mechanism by fusing the binary cross entropy (BCE) and Dice loss functions to alleviate the pixel-level category imbalance problem in CD. They are expressed as follows:

$$\text{Loss} = \text{Loss}_{\text{BCE}} + \text{Loss}_{\text{dice}}. \tag{4}$$

BCE loss is expressed as follows:

$$\text{Loss}_{\text{BCE}}(g, p) = \frac{1}{N}\sum_{i=1}^{N}\left[g_i \log(p_i) + (1 - g_i)\log(1 - p_i)\right]. \tag{5}$$

Dice loss is represented as follows:

$$\text{Loss}_{\text{dice}} = 1 - \frac{2 \cdot \sum_{i=1}^{N} g_i \cdot p_i}{\sum_{i=1}^{N} g_i + \sum_{i=1}^{N} p_i} \tag{6}$$

where g and p represent the ground-truth (GT) label and the predicted result map, respectively. N denotes the total number of pixels. For each pixel i, the true class g_i is 0 for no change and 1 for change. The model predicts the probability that the pixel belongs to the changed class as p, and the probability that it belongs to the unchanged class as $1 - p_i$.

IV. Experiment

This section first introduces two datasets used in the experiments and their configurations. It then details the experimental setup. Subsequently, the performance of the model is evaluated. Finally, the contributions of individual modules and the rationale behind the backbone network are analyzed.

A. Dataset Description

1) LEVIR-CD [38]: The dataset consists of 637 pairs of bitemporal aerial images, with each image measuring 1024 × 1024 pixels and a spatial resolution of 0.5 m. The images are collected by Google Earth. The temporal difference between the images ranges from 5 to 14 years that makes it a high-resolution remote sensing dataset primarily focused on building CD . Due to GPU memory limitations, the original images are split into nonoverlapping patches of 256 × 256 pixels.

2) SYSU-CD [30]: The dataset contains 20 000 pairs of high-resolution remote sensing images from 2007 to 2014. Each image has a size of 256 × 256 pixels at a spatial resolution of 0.5 m. The dataset is released by Sun Yat-sen University. The dataset focuses on detecting various types of land cover changes, including new urban constructions, suburban expansion, preconstruction groundwork, vegetation changes, road extensions, and marine construction.

B. Comparison Methods

To ensure an objective and fair evaluation of the proposed method, this article selects six open-source remote sensing CD networks for comparison with GLONet.

1) STANet (2020) [38]: The network primarily captures spatiotemporal dependencies between pixels through global and multiscale attention modules to improve detection accuracy.

2) SemicdNet (2021) [39]: The method combines a generative adversarial network (GAN) with an attention-enhanced U-Net++ network, leveraging both labeled and unlabeled data to improve detection performance.

3) SNUNet (2022) [10]: The network combines a NestedUNet architecture with an ensemble channel attention module. It utilizes dense skip connections between encoders–decoders and among decoders to maintain high-resolution features and precise localization details.

4) DSAMNet (2022) [30]: The method integrates a deep supervision module with metric learning based on the convolutional block attention module (CBAM). It employs CBAM to enhance feature discriminability and generates CD results by computing feature distance maps through the metric module.

5) GeSANet (2023) [40]: To achieve high-precision image CD, the method corrects image registration offsets through a position matching mechanism and filters out pseudo-change information using a content reasoning mechanism.

6) SFNet(2024) [18]: This network improves remote sensing image CD (RSI-CD) accuracy by integrating an NFAM and a semantic filtering mechanism (MSFM) to enhance the semantic representation and filter pseudo-change.

C. Experimental Setup

1) Evaluation Metrics: To ensure the effectiveness of the CD method, this article adopts six evaluation indicators to assess the proposed approach, including: precision (Pre), recall (Rec), $F1$-score ($F1$), overall accuracy (OA), intersection over union (IoU), and kappa

$$\text{Pre} = \frac{\text{TP}}{(\text{TP} + \text{FP})} \tag{7}$$

$$\text{Rec} = \frac{\text{TP}}{(\text{TP} + \text{FN})} \tag{8}$$

$$\text{F1} = \frac{(2 \times Pr \times Re)}{(Pr + Re)} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{(\text{FP} + \text{TP} + \text{FN})} \tag{10}$$

$$\text{OA} = \frac{(\text{TP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})} \tag{11}$$

$$\text{kappa} = \frac{(\text{OA} - P_e)}{(1 - P_e)} \tag{12}$$

where TP, FN, FP, and TN represent the true positive, false positive, false negative, and true negative, respectively. P_e represents the expected agreement by Chance, which is the theoretical probability of consistency between CD results and reference data due purely to random factors.

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2) Implementation Details: During both the training and testing phases, we employ an RTX 4090D GPU configuration. Each GPU is equipped with 24 GB of memory. We use the AdamW optimizer during training, with an initial learning rate of 0.0001. The learning rate is dynamically reduced by 50% every 30 epochs. The dilation and erosion operations employ a square structuring element of size 3×3 with a stride of 1. In addition, various data augmentation techniques, including random cropping, flipping, and Gaussian noise injection, are applied during data loading to enhance model robustness.

D. Comparisons With the Existing Methods

The evaluation of GLONet includes both quantitative and qualitative experiments conducted on two benchmark datasets, with comparisons made against six existing CD methods. The quantitative assessment employs six standard evaluation metrics, while the qualitative analysis visually examines detection results to demonstrate the effectiveness of model in feature fusion and edge refinement. This combined approach provides a comprehensive validation of the performance of GLONet in addressing key challenges in remote sensing CD.

1) Quantitative Evaluation: According to Table I on the LEVIR dataset, GLONet demonstrates better performance in several metrics, especially in precision and recall. Compared with SFNet, GLONet is more effective in pseudo-change information identification and filtering, significantly reducing the missed alarm rate. In the comparison with SemicdNet, GLONet shows higher recall, which indicates that it is more accurate in handling pseudo-change in complex scenarios. It is noted that GLONet achieves a positive balance between recall and precision. It avoids the issue of increased pseudo-change caused by excessive filtering, as seen in STANet. Compared with other methods, GLONet maintains high detection accuracy while reducing pseudo-variation interference and performs more stably, especially in high-complexity scenarios. Compared with GeSANet, GLONet not only demonstrates the necessity of multiscale feature exploration but also further improves the filtering effect of pseudo-change through its innovative feature association mechanism. Ultimately, the final quantitative evaluation results demonstrate that GLONet outperforms the comparative methods across all six evaluation metrics. It proves its superiority and practicality on the LEVIR dataset.

As shown in Table I, GLONet demonstrates better overall performance compared with other algorithms on the SYSU dataset. Particularly when compared with SFNet, it shows significant advantages in precision metrics and exhibits more outstanding detection capabilities in complex scenarios. GeSANet maintains a more robust network performance in complex scenarios. Its ranking behind our algorithms proves the necessity of multiscale feature exploration and pseudo-variation filtering. In contrast, SemicdNet exhibits better precision. The overall performance is more consistent on the SYSU dataset, and its generalization ability in complex scenarios is demonstrated. However, the performance of STANet is more unstable, and there is a significant gap between recall and precision, which reflects the leakage detection problem caused by over-filtering SNUNet, although it does not show performance imbalance, still fails to achieve the expected results. In summary, GLONet can effectively reduce the pseudo-change rate and leakage rate in dealing with complex scenarios and thus improves the overall detection performance, especially showing advantages in $F1$ and IoU.

2) Qualitative Evaluation: The visualization results in Fig. 4(a)–(j) demonstrate that in complex scenarios involving roads, illumination, vegetation, and seasonal variations, existing methods (e.g., STANet and SemiedNet) exhibit noticeable false detections (blue). These methods also show missed detections (red) under such challenging conditions. These issues arise due to insufficient feature extraction capability and limited pseudo-change filtering effectiveness. For instance, STANet causes broken road edges due to over-filtering, while SemiedNet misidentifies vegetation shadows as changes under seasonal illumination interference. The proposed GLONet significantly reduces pseudo-change interference (e.g., from illumination and shadows) through its LGDFM, which dynamically integrates multiscale features. In addition, it employs the EAEM for precise contour extraction. Simultaneously, it completely preserves the edge structures of genuinely changed targets (e.g., buildings and roads), with a final prediction result that is closer to the GT.

Compared with Fig. 4, the changes in Fig. 5 primarily focus on building clusters, vegetation coverage, and surface water bodies. The visualization results show that all comparative methods exhibit varying degrees of pseudo-change (marked in blue) and missed detections (marked in red) on the SYSU dataset. In the third row of Fig. 5(d)–(i), STANet and SemiedNet suffer from severe missed detections at building edges due to over-filtering, while SNUNet shows extensive pseudo-change interference in vegetation areas. In contrast, DSAMNet and GeSANet significantly reduce false detections in road change areas, while their performance in detecting small water body changes remains limited. Our proposed GLONet [see Fig. 5(i)] not only accurately identifies newly constructed buildings (marked in white) but also achieves complete detection of road expansions and water boundary changes. Its prediction results demonstrate optimal spatial consistency and detail integrity compared to the GT. This validates the effectiveness of the LGDFM in spatiotemporal feature fusion and the EAEM in detail preservation.

E. Ablation Study

To rigorously evaluate the contributions of each component in GLONet, we conduct comprehensive ablation studies on two public datasets, as detailed in Table II. The baseline model incorporates the SAFG and MSFA modules, which serve as foundational elements but are not the primary innovations of this work. Building upon this baseline, Network1 integrates the proposed LGDFM to assess its impact on feature fusion and change representation. Network2 further extends the baseline by introducing the EAEM to evaluate its role in edge refinement and detail preservation. Finally, Network3 represents the complete GLONet framework, combining both LGDFM and EAEM to demonstrate their synergistic effects on overall performance. This systematic ablation analysis validates the necessity and effectiveness of each proposed

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TABLE I

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE LEVIR AND SYSU DATASETS. THE BEST SCORES ARE IN BOLD, AND THE SECOND HIGHEST SCORE IS UNDERLINED

Dataset	Method	Pre(%)	Rec(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)
LEVIR	STANet	73.80	82.52	77.93	92.22	63.37	71.85
	SemicdNet	90.48	64.86	72.65	93.37	61.07	69.40
	SNUNet	90.28	88.93	89.60	96.86	81.62	87.31
	DSAMNet	89.89	84.89	87.31	96.13	77.61	84.17
	GeSANet	<u>90.74</u>	88.76	89.73	96.91	81.32	87.49
	SFNet	90.64	<u>89.59</u>	<u>90.11</u>	<u>96.96</u>	<u>82.25</u>	<u>87.98</u>
	GLONet(Ours)	**92.10**	**91.06**	**91.58**	**99.14**	**84.47**	**91.13**
SYSU	STANet	40.90	68.89	50.47	69.50	33.75	30.14
	SemicdNet	79.62	60.73	68.91	87.07	52.56	60.93
	SNUNet	63.68	61.30	62.46	82.63	45.42	51.17
	DSAMNet	78.06	60.90	68.42	86.74	52.00	60.19
	GeSANet	79.08	72.09	75.43	88.92	60.55	68.29
	SFNet	<u>80.52</u>	**75.54**	<u>77.97</u>	<u>89.92</u>	<u>63.87</u>	<u>71.43</u>
	GLONet(Ours)	**86.67**	<u>75.28</u>	**80.70**	**91.84**	**68.07**	**76.47**

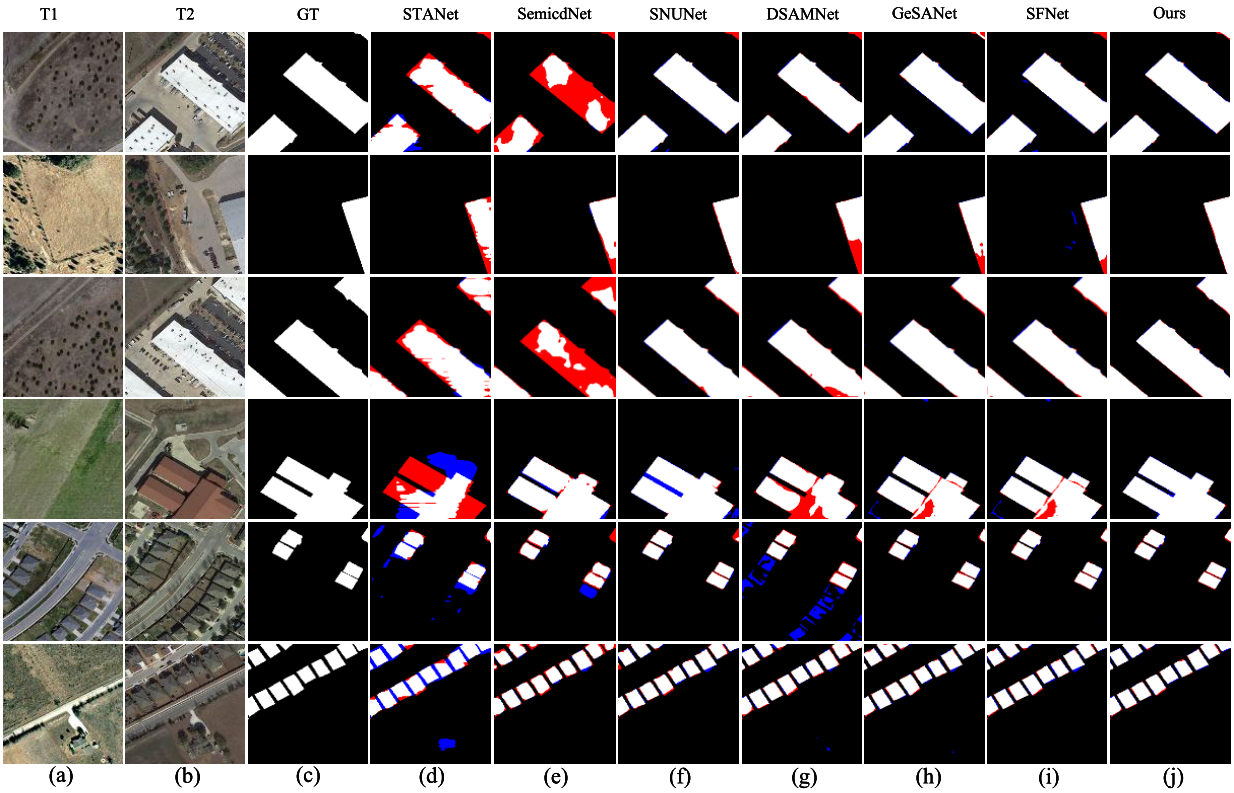


Fig. 4. Visualization results of different methods on the LEVIR dataset. Blue represents the pseudo-change, red indicates miss detection, and white denotes correct detection. (a) T1. (b) T2. (c) GT. (d) STANet. (e) SemicdNet. (f) SNUNet. (g) DSAMNet. (h) GeSANet. (i) SFNet. (j) Ours.

module in addressing the challenges of bitemporal feature fusion and edge-aware CD.

1) Study of LGDFM: The core advantage of the LGDFM lies in its local–global feature cooperative differential fusion mechanism. We can analyze its effectiveness from several key aspects. First, the LGDFM uses stacked convolutional layers to progressively extract high-resolution local features such as building edges and road textures, preventing the spatial information loss that occurs with traditional CNN downsampling. Second, the module integrates a self-attention mechanism that computes pixel-level global relationships, which solves the incomplete detection problem for large-scale changes like newly constructed urban areas in conventional CNNs. Third, the module applies Sigmoid to generate channel weights, enhancing real change regions, including new buildings, while suppressing pseudo-change, such as shadows and seasonal variations. Experimental results demonstrate the capabilities of the module. The LGDFM enhances precision on the LEVIR dataset by reducing false positives resulting from edge ambiguity through its fine-grained features. For the SYSU dataset, the module increases recall from 73.58% to 74.10% by capturing scattered changes like vegetation shifts and water expansion through global modeling, reducing missed detections. In addition, the 90.81% $F1$-score gain on the LEVIR dataset proves the module effectively balances precision and recall via adaptive weight allocation, avoiding bias toward either metric. In summary, the LGDFM combines multiscale feature fusion with dynamic differential strategies to maintain strong

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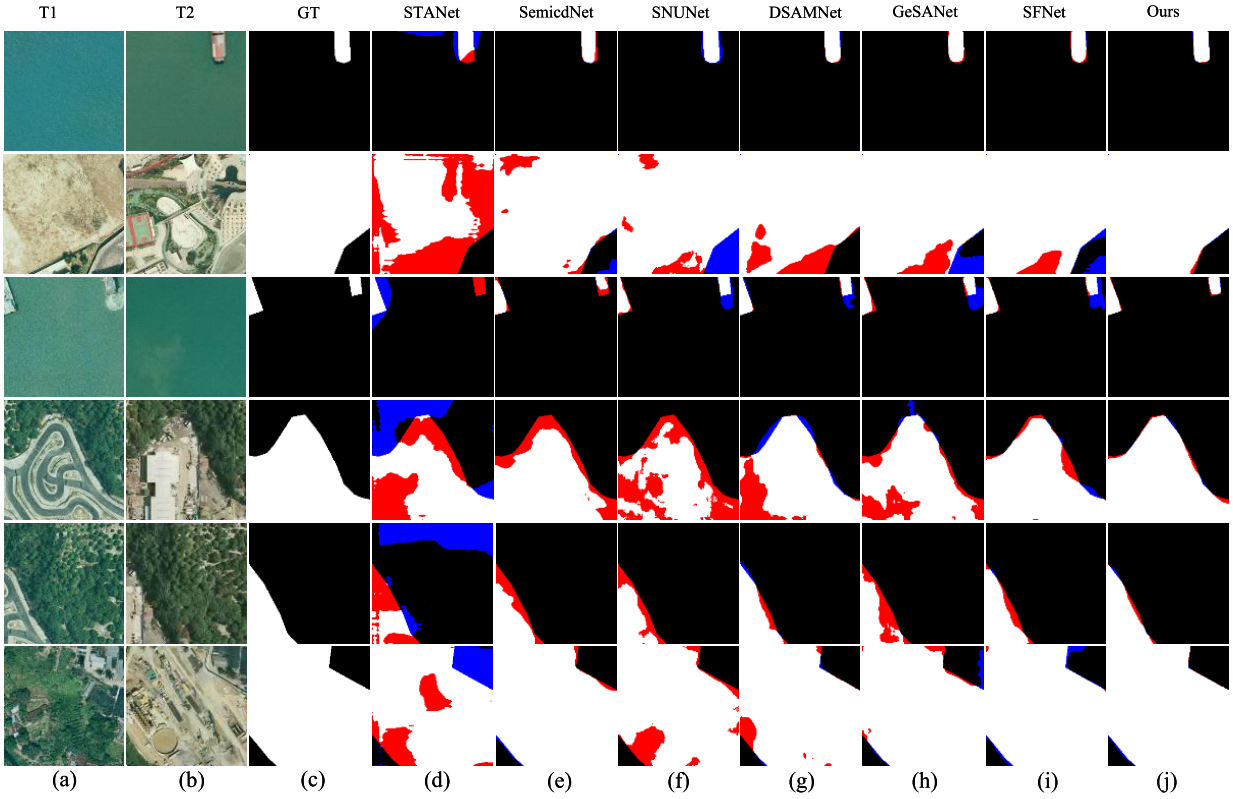


Fig. 5. Visualization results of different methods on the SYSU dataset. Blue represents the pseudo-change, red indicates miss detection, and white denotes correct detection. (a) T1. (b) T2. (c) GT. (d) STANet. (e) SemicdNet. (f) SNUNet. (g) DSAMNet. (h) GeSANet. (i) SFNet. (j) SFNet.

robustness even under challenging conditions like illumination changes and registration misalignment.

2) Study of EAEM: The effectiveness of the EAEM is primarily reflected in its unique edge-aware mechanism and optimization strategy. We can conduct an in-depth analysis from the following aspects. First, the module employs morphological operations (dilation–erosion) to explicitly extract edge features. By adjusting the kernel size of dilation or erosion, it can flexibly adapt to edge features at different scales. Second, the module achieves adaptive enhancement of edge information by multiplying the edge features E with the original features F at the pixel level and then summing them. This operation essentially applies spatial attention weighting to the feature maps, enabling the network to focus on boundary regions. This is the key reason for the 0.5% improvement in the IoU metric. Finally, the EAEM introduces a Transformer layer for global optimization. The long-distance dependency characterization capability of the Transformer helps correct geometric distortions caused by the local processing of CNNs. By incorporating edge information at different stages, the module achieves multiscale feature integration from fine-grained edges to global semantics. This is a major factor in the kappa coefficient increasing from 90.29% to 90.61%. Experiments demonstrate that on the LEVIR dataset, precise edge localization effectively distinguishes real building changes from pseudo-changes such as shadows. This explains why Pre improves from 91.62% to 92.01%. On the SYSU dataset, enhanced processing of irregular edges (e.g., vegetation cover changes and water boundaries) increases recall from 73.58% to 74.62%. In summary, the EAEM significantly improves the completeness of edge details while maintaining high accuracy through its edge extraction and optimization mechanisms.

3) Study of GLONet: From the ablation experiments in Table II, it can be observed that the complete GLONet achieves better performance on both datasets. First, the LGDFM is one of the core innovations of GLONet. Its effectiveness is demonstrated by capturing fine-grained features through the local feature extraction module and modeling pixel-level global relationships through the global feature extraction module (self-attention mechanism), addressing the challenge of large-scale CD. The difference strategy employs the Sigmoid function to generate channel weights, which enhances real change regions while suppressing pseudo-change. Second, EAEM significantly improves the contour accuracy of change targets through its unique edge-processing mechanism. The introduction of a Transformer layer for global optimization leverages its long-range dependency modeling capability to correct geometric distortions caused by the local processing of CNNs. Finally, the performance of the complete GLONet (Network3) is not simply the sum of the effects of individual modules but rather a result of their synergistic interaction. The high-quality difference features provided by the LGDFM lay a solid foundation for edge extraction in the EAEM, while the edge information extracted by the EAEM in turn guides the LGDFM to focus more on real change boundaries. This interaction attains an $F1$ of 91.58% on the LEVIR dataset, surpassing networks that use either module alone. In summary, through LGDFM and EAEM, GLONet effectively alleviates two key challenges in remote sensing CD: insufficient fusion of bitemporal features and imprecise edge detection.

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TABLE II
EVALUATION METRICS COMPARISON OF ABLATION EXPERIMENTS. THE USE OF MODULES OR NOT IS REPRESENTED BY ✓ AND ×

Dataset	Model	LGDFM	EAEM	SAFG	MSFA	Pre	Rec	F1	OA	IoU	Kappa
LEVIR	Baseline	×	×	✓	✓	91.62	88.06	90.67	99.02	83.11	90.29
	Network1	✓	×	✓	✓	91.70	88.99	90.81	99.08	83.17	90.33
	Network2	×	✓	✓	✓	92.01	88.31	91.07	99.12	83.61	90.61
	Network3	✓	✓	✓	✓	92.10	91.06	91.58	99.14	84.47	91.13
SYSU	Baseline	×	×	✓	✓	85.26	73.58	80.57	90.98	66.81	75.36
	Network1	✓	×	✓	✓	86.65	74.10	80.13	91.45	66.84	74.74
	Network2	×	✓	✓	✓	85.42	74.62	80.64	90.96	67.30	75.82
	Network3	✓	✓	✓	✓	86.67	75.28	80.70	91.84	68.07	76.47

TABLE III
PERFORMANCE COMPARISON OF CD BETWEEN RESNET-18 AND RESNET-34

Dataset	Backbone	Pre	Rec	F1	OA	IoU	Kappa
LEVIR	ResNet-34	92.16	90.98	91.51	99.11	84.39	91.05
	ResNet-18	92.10	91.06	91.58	99.14	84.47	91.13
SYSU	ResNet-34	86.71	75.11	80.62	91.80	67.92	76.41
	ResNet-18	86.67	75.28	80.70	91.84	68.07	76.47

4) Study of Backbone Network: In GLONet, ResNet-18 outperforms ResNet-34 primarily due to its lightweight design, which is better suited for high-resolution image processing as shown in Table III. This avoids the parameter redundancy and overfitting risks associated with deeper networks. The LGDFM enhances global modeling capabilities through self-attention mechanisms. This compensates for the limited receptive field of CNNs. In this context, the deep global modeling in ResNet-34 may cause functional redundancy. In comparison, the moderate depth of ResNet-18 shows better compatibility with modular task division. In addition, the EAEM relies on low-level edge features extracted by CNNs. The shallower architecture of ResNet-18 may preserve more low-level details, while the higher level features of ResNet-34 could become overly abstract, hindering edge refinement. On the LEVIR dataset, ResNet-18 attains higher $F1$ and IoU scores. On the SYSU dataset, it demonstrates superior kappa values. This indicates that ResNet-18 holds an advantage in task-critical metrics such as pseudo-change filtering and edge preservation.

V. CONCLUSION

In this article, we propose GLONet, an innovative framework for remote sensing image CD, which addresses the critical challenges of inadequate bitemporal feature fusion and imprecise edge detection. The local–global differential fusion module effectively integrates multiscale features through adaptive differential fusion, while the EAEM enhances contour accuracy using morphological operations and global optimization. Experimental results on the LEVIR-CD and SYSU-CD datasets demonstrate that GLONet outperforms existing methods in both quantitative metrics and qualitative performance, achieving superior precision, recall, and edge preservation. Future work will focus on developing lightweight architectures to improve computational efficiency and exploring advanced feature interaction mechanisms for large-scale applications. In addition, extending GLONet to multitemporal CD and integrating explainable AI techniques could further enhance its practicality and interpretability in real-world scenarios.

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